

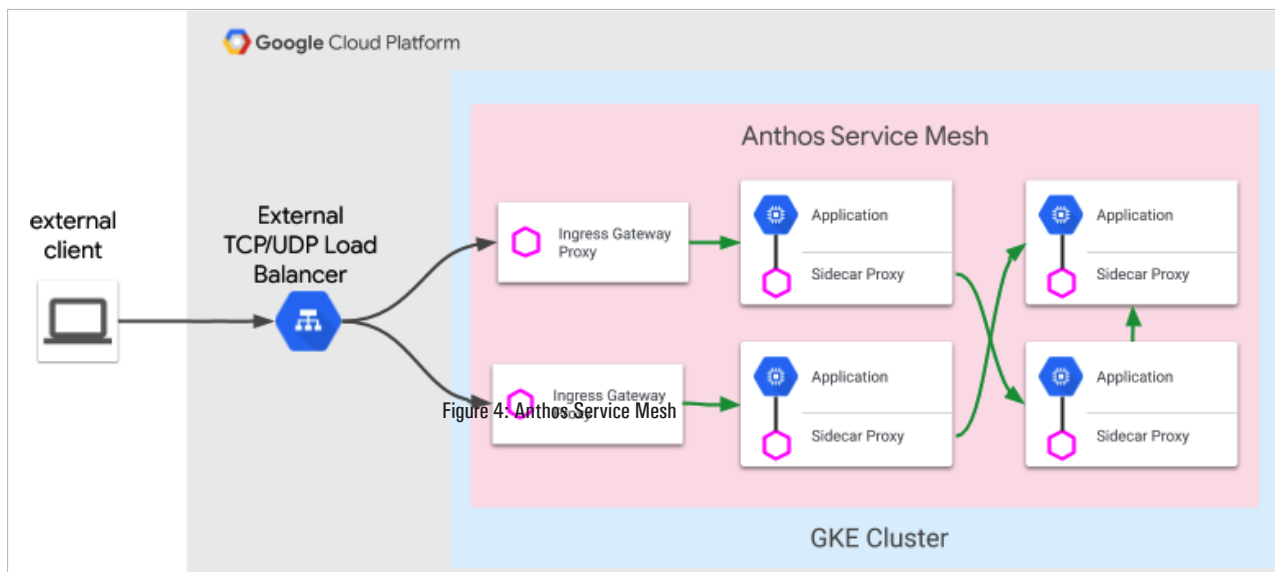


Figure 4: Anthos Service Mesh

inter-service communication. When dealing with a large number of microservices/APIs with extensive inter-service communications, service mesh architecture is suitable for better resource handling and communication patterns.

API gateway manages external traffic from service clients to internal resources such as APIs/microservices. It enables routing to API services and exposes business functions to the external world. When you have many service client calls (request processing) to API services, you can choose an API gateway to handle the traffic efficiently, along with response caching and internal load balancing, ensuring high-availability without compromising security and performance.

Anthos service mesh

Google's native service mesh feature is available as Anthos service mesh, featuring one or more control planes (fully-managed services that can be configured with Google handling patch/upgrade, scalability, and security features) and a data plane handling workload instance communication activities.

Anthos Service Mesh is based on the powerful Istio framework

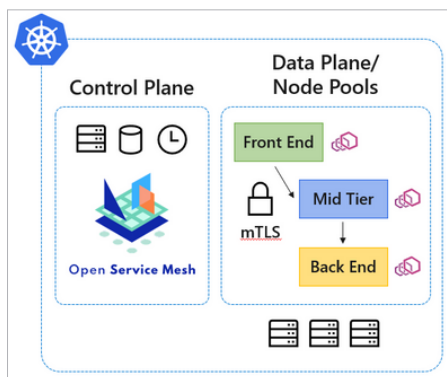


Figure 5: Open Service Mesh

and the control plane is based on a consolidated monolithic binary called Istiod. It uses a proxy to monitor all traffic in service communications. The service mesh provides fine-grained control over service communications and proxy services, and load-balancing between services. It controls service traffic flows in both directions, such as traffic into the mesh/services (ingress traffic) and outside services (egress traffic management).

Anthos Service Mesh is integrated with cloud monitoring, cloud logging, and cloud trace for native integration, offering observability insights. It employs higher security features like mutual transport layer security (mTLS) for authenticating service peers. It

can manage managed instance groups (MIGs) of compute engine instances and Google Kubernetes Engine (GKE) clusters within the service mesh for reliable proxied communication.

Open Service Mesh (OSM) add-on to enhance AKS features

Microsoft initially developed Open Service Mesh (OSM) for fully-managed service mesh

features and later donated it to the Cloud Native Computing Foundation (CNCF) as an open source project. It serves as an envoy-based control plane on the Kubernetes engine, managing and configuring infrastructure services on-the-fly.

OSM was included as an add-on to Azure Kubernetes Service (AKS) as a lightweight service mesh feature for Kubernetes applications. It provides access control policies, monitoring, and debugging for these applications. In March 2021, the OSM add-on was announced as a public feature or general availability in AKS, enabling service-to-service communication in the Azure platform using mTLS, facilitating two-way encrypted TLS communication between services.